SoK: Evaluating LLM and Agent Side-Channel Defenses From Evidence Audits to Real-Agent Measurements

***Abstract*—Encryption hides message content while leaving the structure of agent workflows observable. Determining whether a side-channel defense protects a deployed workflow requires connecting the evidence across observation, execution and evaluation boundaries. We systematize 82 study families (58 core and 24 adjacent), with 39 scoped result records and an audit of 20 mitigation families. Our nine-item audit and three deployment checks identify concrete gaps: only 3 of 17 applicable families bind task or component utility, and 2 of 11 bind a matched adaptation contrast. Controlled experiments show why these bindings matter: per-event padding leaves streaming verbosity fully separable, refitting raises attack accuracy from 50% to 100%, and encrypted request sizes identify retail tools with 89.9–91.4% held-out accuracy. We then evaluate 690 real GPT-6.1 Sol agent runs, calibrating the defense solely on 414 training runs before collecting 276 customer-disjoint test runs. In local TLS replay, online shadow cover with constant framing reduces cancellation/return network balanced accuracy from 97.2%/94.0% to 50.0%/50.0%, at 15.4× bytes and 3.8× calls. Workflow rules attain 97.1% measured compliance while intent inference remains above 93%. The intent signal lives in state-changing tool calls, which alone support 84–92% accuracy; the rules standardize only the reads. Customer-grouped cross-fitting over all 690 runs confirms a 31–39-point reduction. It also identifies a bucketed operating point that reaches 52.2%/52.7% at 12.4× bytes, 37% cheaper than constant framing at matching or lower leakage. The resulting evidence framework connects source-level audit findings to reproducible measurements of leakage, state preservation and deployment cost.**

1. Introduction

Encryption conceals message content, yet LLM serving engines and agent systems expose packet sizes, cache timing, memory-access patterns, visited domains and tool-invocation sequences [1], [2], [3], [4], [5], [6]. A published defense result applies to a particular observer and operating point. Reusing it across a frontend, a tool dispatcher or another workload can change both the observable signal and the cost being measured. We turn this deployment problem into explicit, testable requirements for reusing defense evidence.

Our systematization connects published results to the receiving deployment through three operational checks:

1) **Bind the receiving projection.** Target-oblivious postprocessing is a sufficient condition for preserving an upstream information bound (Remark 1, a direct consequence of the data processing inequality). A frontend with access to decoded content or an unprotected backward computation requires a separate argument or measurement [7], [8].
2) **Bind execution and protected state.** Read-only decoy requests and GUI actions operate on different interfaces and serve different objectives [6], [9]. For tools, unchanged argument values do not imply unchanged database state. The relevant contract must identify task objects, the user's account and shared state separately.
3) **Bind adaptation, utility and cost.** An attack on defended data must specify how the attacker was fitted; an overhead must identify the same workload and deployed mechanism as the claimed protection. Binding these quantities makes protection and cost comparable across deployments.

Contributions. We connect a source-bound systematization to executable tests of defense reuse across observation and execution boundaries. The result is a measurement framework with three concrete outputs:

- **A result-record repository:** 82 study families (58 core and 24 adjacent), with 39 scoped records and 78 utility/cost field positions from 26 families. The records distinguish source versions, receiver views, execution modes and cost objects that paper-level comparisons can conflate.
- **An actionable comparability audit:** nine reporting items and three operational checks applied to 20 mitigation families. Utility is bound in 3/17 applicable families and a matched adaptation contrast in 2/11. The audit maps each unresolved reuse claim to the specific observation, state or evaluation evidence needed to resolve it.
- **An executable bridge to real-agent defense evaluation:** controlled streaming and tool-state experiments culminate in 690 real-agent runs, with 414 training runs for defense calibration and 276 customer-disjoint test runs. Joint online cover and framing reduce cancellation/return network BA by 47.2/44.0 percentage points. A rules-prompt control reaches 97.1% measured compliance while retaining over 93% intent inference, separating behavioral compliance from metadata protection. Leakage localization explains why: state-changing calls alone support 84–92% balanced accuracy, and the rules

leave them unchanged. Cross-fitting over all 690 runs confirms the defense effect (31–39 points) and yields a bucketed operating point at 12.4× bytes, 37% below constant framing.

The contribution is the connection between evidence and execution: result-level records expose ambiguous comparisons, operational checks specify the missing bindings, and controlled and real-agent measurements demonstrate their consequences. Composition, state isolation and adaptive evaluation provide the foundations for this reproducible assessment.

Table 7 links source-local findings to the new comparison judgments. For example, separating Dangol's stored-capture and live-protocol records prevents an invalid subtraction; separating Early Bird's two mechanisms prevents using a semantic-cache cost to price KV-prefix isolation. The transfer tests add consequences that a paper-level reading misses: a buffered padding cost does not price a stream, and leakage already present at a dispatcher persists through size-based decoding on the encrypted tool link.

Figure 1 depicts the trace and action paths; Section 8 tests each path and their composition on one agent–tool link. Figure 2 summarizes the checks.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and when a component intervention protects an end-to-end workflow at a bounded utility and cost (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [10], [11], [12], [13], [14]. Table 1 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest:

1) **Type I (Confidentiality and privacy protection):** An unauthorized observer attempts to infer private user prompts, personal traits, visited domains or proprietary model states; the defense protects user or tenant confidentiality (e.g., NetEcho, WRA, Early Bird, KVGov, CachePrune, OTRO [7], [6], [3], [15], [16], [17]).
2) **Type II (Provenance and identity attribution):** An observer infers which model, agent harness or user identity generated a trace from timing, packet or DOM event structure (e.g., UI Traces, FP-Agent, AgentPrint, Lidar [5], [18], [4], [19]).
3) **Type III (Human–agent / bot-detection evasion):** The receiving platform operates a defensive classifier to detect automated bots, while the agent applies "action humanization" (e.g., AHB [9]) to evade detection—inverting the defender/attacker polarity relative to Type I privacy protection.
4) **Type IV (Authorized operator diagnostics):** Privileged loop detection, serving root-cause localization or internal residual-stream monitoring (ClawGuard, cycle detection, LumiMAS, LLM RCA, MIRAGE [20], [21], [22], [23], [24]), which use C1–C5 auxiliary signals for legitimate system telemetry rather than privacy exploitation.

TABLE 1. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Seed-anchored purposive corpus and exact reconciliation

Because LLM and agent side-channel papers are scattered across security conferences, systems venues and preprints under heterogeneous terminology ("remote keylogging," "prompt-cache leakage," "workflow fingerprinting," "agent humanization"), our corpus is constructed through *purposive seed selection, venue-targeted proceedings screening (IEEE S&P, USENIX Security, NDSS, ACM CCS, PMLR) and one-hop backward/forward citation snowballing* from 40 fixed seeds (2,237 backward reference occurrences and 24 distinct forward works screened; 38/40 forward seeds resolved; seed resolution and the number of

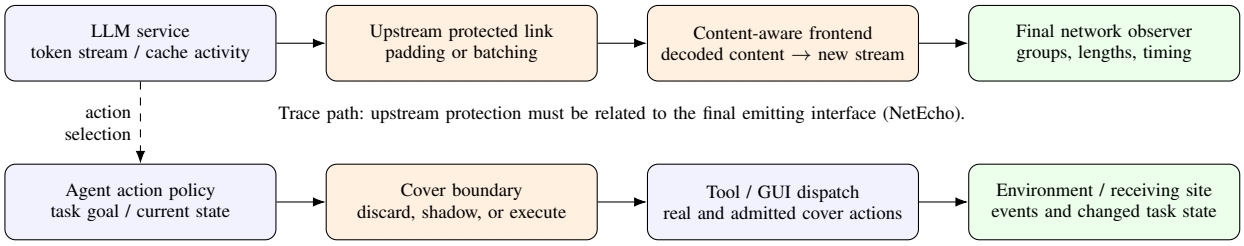


Figure 1. Observation and execution boundaries represented by a result record. E5 tests the trace path, E6 the action path, E7 their encrypted composition, and E9 real-agent trajectories with training-only calibration.

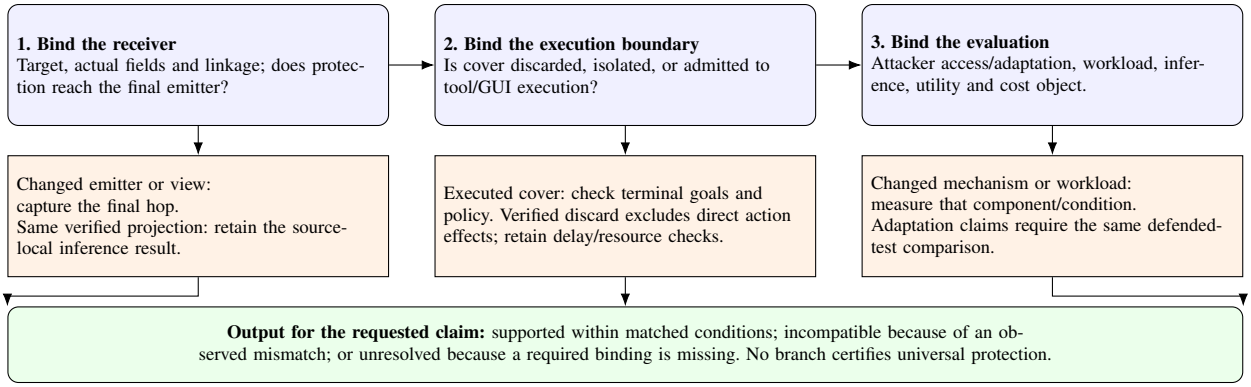


Figure 2. Three checks for deciding which evidence can be reused in a target deployment.

distinct returned works are different counts), complemented by native database exports and bulk keyword searches (Appendix A and Supplement S1/S10).

To document why standalone Boolean keyword queries cannot replace seed-and-snowball discovery, we audited both our historical October 8–9 native IEEE/ACM exports and a completed October 21 supplementary search frame comprising four Semantic Scholar title/abstract bulk queries and two native arXiv searches (615 occurrences deduplicated to 565 unique records; Table 2). Those six supplementary keyword queries returned 17/24 of a fixed 24-paper core subset of our 40 seeds (`seed_n=24`) and 25/60 of the October 8 snapshot. Because the 24 known papers are a core subset of the 40 initial seeds rather than an external holdout, 17/24 and 25/60 measure *lexical overlap between narrow Boolean queries and our seed-anchored corpus*, not calibrated database recall: the two native arXiv queries required the exact hyphenated string `"side-channel"` (missing studies titled with "prompt leakage," "keylogging," "timing attacks" or "fingerprinting"), and Semantic Scholar title/abstract matching did not search full texts of USENIX/NDSS PDFs.

Table 2 reconciles every record across the 60 → 82 transition. All 58 core families (20 workflow-origin, 38 inherited-runtime/application; 53 non-diagnostic and 5 Type IV diagnostic) and 24 adjacent families are integrated into the unified corpus maps and intervention dispositions (Supplement S2/S5). Across the 58 core families, 20 evaluate mitigations (16 snapshot families plus four additions). The result ledger is broader: 33 baseline records from 22 families, including six contextual comparators, plus six records from the four added mitigation families. Thus the 39-record ledger spans 26 families and 78 utility/cost field positions; the reporting-audit denominator is 20.

2.3. Supplied coding records and agreement

The supplied package records a primary AI-assisted coding workflow and a second coding column described by its provenance as a domain-expert review. We retain the recorded agreement summaries and supply their source columns. The files alone do not verify coder identity, blinding or independence from the primary decisions; we therefore report annotation agreement rather than newly verified independent human validation. The recorded scope is: The saved screening sample has 93/100 three-class agreement ($\kappa = 0.886$); the nine-item baseline audit has 134/144 cell agreement across 16 families ($\kappa = 0.876$); and the 50-pair sample has 141/150 judgment agreement ($\kappa = 0.916$). The latter covers all 13 same-family pairs plus 37 other within-cohort pairs. Full intervals and source locators remain in Supplement S8 and the annotation files.

The historical confidence intervals resample individual screening records, audit cells and pairwise judgments respectively. Cells within families and three judgments within a pair are dependent, so these intervals are conditional

TABLE 2. ITEM-BY-ITEM RECONCILIATION OF THE 565-RECORD OCTOBER 21 SEARCH FRAME (BEFORE AND AFTER THE 204-RECORD EXCLUSION AUDIT) AND THE TRANSITION FROM THE 60-FAMILY SNAPSHOT TO ALL 82 CODED FAMILIES.

Route / family category	Pre-audit	Audit Δ	Final (Indexed)
Excluded: no hidden C2/C3 target	221	−4	217 (0)
Excluded: outside LLM / diff. target	175	0	175 (0)
Existing snapshot core / adjacent	23 / 1	+1/0	24 / 1 (in 60)
New source-verified core families	14	+1	15 (**+15**)
Adjacent source-reviewed contexts	28	+2	30 (**+7**)
Adjacent metadata-only contexts	28	0	28 (0)
Background / proposal metadata	58	0	58 (0)
Missing abstract / unavailable text	17	0	17 (0)
565-frame total / Net added families	**565**	**0 net**	**565 (+22)**
Coded collection: Core / Adjacent (**Total**)	Snapshot: 43 / 17 (**60**) → Full: 58 / 24 (**82**)		

The exclusion audit selected 177 high-risk pattern matches plus 27 reason-stratified records (seed 20261021), rather than all 396 initially excluded records. This enriched 204-record subset rerouted 4 records out of the 221 initial C2/C3 exclusions: (1) `SUP-0192` (MIRAGE [24]) moved to new core ($14 + 1 = 15$, all 15 added to the family index); (2) `SUP-0552` (InputSnatch arXiv alias) merged into existing family `inputsnatch2024` ($23 + 1 = 24$, +0 new families); and (3–4) `SUP-0191` and `SUP-0424` moved to source-reviewed adjacent contexts ($28 + 2 = 30$, of which 7 boundary-defining studies enter the adjacent index and 23 remain in the 565-record ledger). Thus $60 + 15$ (core) $+ 7$ (adjacent) $= 82$.

descriptive summaries, not cluster-robust generalization intervals. The 36 cells for four added families are outside the 144-cell agreement calculation. Agreement also does not establish coding correctness, and κ depends on category prevalence.

SoK Takeaway 1 (Search and evidence scope). The supplementary queries returned 25/60 snapshot families and 17/24 known seed papers. These overlaps motivate complementary discovery methods; they neither estimate field-wide recall nor identify why each missing paper was absent. Objective, observer and authorization must be specified separately when comparing interventions.

3. Result Records, Comparability Rules and Validity

The atomic unit of our repository is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0 vs. D_2), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [25]), distinct

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric/denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [26] and OTRO [17]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (NR-R), unavailable (NR-A) or unverified (NR-V). Table 5 illustrates a complete record-to-judgment lineage.

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	`dangol2026workflow` / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [25].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

TABLE 6. AUTHOR-CONSTRUCTED SCENARIO EXERCISE: COMPLETE CONFUSION MATRICES. EACH TRIPLE GIVES PREDICTIONS (COMPATIBLE, INCOMPATIBLE, UNRESOLVED) FOR THE INDICATED TRUE CATEGORY. THE LAST TWO COLUMNS DISTINGUISH DETECTING MISMATCHES FROM RETAINING COMPATIBLE REUSE.

Procedure	True C	True I	True U	I recall	C recall
Always C	7,0,0	12,0,0	5,0,0	0.0	100.0
Always I	0,7,0	0,12,0	0,5,0	100.0	0.0
Always U	0,0,7	0,0,12	0,0,5	0.0	0.0
Coder column 1	7,0,0	0,12,0	0,1,4	100.0	100.0
Coder column 2	7,0,0	0,11,1	0,1,4	91.7	100.0

This illustrates the recorded labeling exercise; scenarios and labels were not independently held out. C recall is specificity when Incompatible is the positive class and evaluation is restricted to the C/I subset. Unresolved is retained as a third outcome.

Illustration rather than external validation. We retain a 24-scenario exercise constructed by the authors from the same corpus used to develop the checks: seven Compatible, twelve Incompatible and five Unresolved cases. The Incompatible category includes proposed transfers with established record mismatches; these are not twelve independently observed deployment failures. The supplied annotation file agrees with the author labels on 23/24 and 22/24 cases for its two coder columns. Agreement between these columns is 21/24 ($\kappa = 0.795$). These descriptive results do not establish framework accuracy on unseen deployments, independent ground truth or superiority over other reading procedures.

A rule that predicts Compatible for every case scores 7/24, a class-count consequence. Always predicting Incompatible instead gives 12/24, 100% recall for that category and 0% recognition of compatible cases. Table 6 reports all outcomes rather than a headline accuracy advantage. A fair comparison would supply the same source information to competing procedures and use independently constructed held-out scenarios; that experiment has not been performed.

The synthetic indexing comparison (Supplement S7) likewise does not establish effectiveness of the three checks. Its 512 roots each contain three query judgments; thus 695 and 476 decision-correct/count-incorrect judgments are out of 1,536 queries per arm. The joint endpoint favors the structured checklist over raw SCOPE traversal. We use SCOPE only as a schema for locating evidence.

4. Corpus Map Across 82 Study Families

4.1. Origins, receiving projections and threat polarities

Across all 58 core families in our unified repository, 38 study inherited runtime, serving or hardware channels (Origin B) and 20 study agent-workflow channels arising from orchestration, tool dispatch, browser navigation and multi-agent protocols (Origin A; Supplement S2 (Table S3)). Signal generation origin is distinct from both receiver location and threat polarity.

Type I: Confidentiality and privacy channels. Inherited serving and transport attacks infer confidential prompts or topics from packet lengths, inter-packet timing, KV-cache sharing and hardware contention. Weiss reconstructs token sequences from encrypted packet lengths, whereas NetEcho recovers conversation semantics after application-layer rechunking [1], [7]. Whisper Leak classifies prompt topics from streaming traffic; Time Will Tell uses total completion duration [27], [28]. Chosen-request cache attacks probe shared KV prefixes or semantic caches via timing oracles, while CPU-cache, GPU memory-barrier, attention-sparsity and page-migration attacks exploit co-resident hardware channels [2], [3], [29], [30], [31], [32], [33], [34], [35], [36], [37]. In split learning/inference, SplitDecoy shows that even when forward activations are masked with decoys, the zero/nonzero row support of returned backward gradients exposes real token positions [8]. At the workflow layer, WRA infers private user functional intent and personal traits from visited domain sequences [6], AgentTell uses probe websites to extract behavioural secrets [38], and routing-metadata attacks infer sensitive prompt topics from model/expert routing choices [39].

Type II: Provenance, model and identity attribution. Type II groups studies by an attribution target: *which model, agent harness or user* produced a trace; such attribution can also raise Type I confidentiality concerns. UI Traces attributes browser-agent identity across 14 models from own-site DOM/JS event sequences [5]; FP-Agent combines behavioural traces with static browser attributes [18]; AgentPrint, Whose Agent Are You?, LLM-WFIN and the cloud traffic benchmark classify workflows or model families from network flows [4], [40], [41], [42]. AgentProv audits tool-use provenance, Lidar fingerprints agent harnesses from action roles and timing, and Cross-Campaign Attribution links asynchronous attack agents [43], [19], [44].

Type III (Bot-detection evasion) and Type IV (Authorized diagnostics). Type III studies evaluate human–agent distinction [45], [9]. Here the threat polarity is inverted

relative to Type I privacy: in AHB [9], the platform detector is a defensive anti-automation control, while "action humanization" is an evasion mechanism applied by the agent. Finally, five core families are Type IV authorized operator diagnostics: ClawGuard, cycle detection, LumiMAS, LLM-serving root-cause analysis (120 controlled fault injections) and MIRAGE (internal residual-stream hooks) [20], [21], [22], [23], [24].

4.2. Relationship to prior systematizations

Prior surveys establish compound-system threat models and confidential-computing boundaries. Banerjee et al. categorize cross-layer compound AI threats under MITRE tactics; Forough et al. distinguish TEE isolation boundaries from side-channel resistance and note that inference protections need not cover agent memory or action coordination [46], [47]. Kotevska et al. propose a six-element privacy contract (asset, observer, channel, permitted disclosure, guarantee, evidence) with composition conditions framed as formal conjectures [48], while classical website-fingerprinting works establish bandwidth/latency bounds for transport shaping (Tamaraw, WTF-PAD, FRONT, Walkie-Talkie) [49], [50], [51], [52]. Our distinct contribution is the populated 82-family / 39-record repository, the nine-item comparability audit, and pre-registered tests of the transfers it flags, using loopback TLS replays and a pinned stateful tool simulator. Website-fingerprinting defenses already show that per-packet padding leaves counts and that constant-shape shaping is expensive [49], [50]. E5 measures how that known property invalidates a specific cost transfer for LLM token streams, without claiming it as new.

5. Three Literature Case Studies

Applying our repository schema to the literature resolves apparent contradictions across three fundamental deployment decisions (Table 7): (1) when upstream traffic shaping survives downstream workflow hops; (2) why read-only query decoys and live GUI humanization produce opposite task outcomes; and (3) how adaptive retraining (D_2) and mechanism-specific cost boundaries govern operating-point selection.

5.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. When an upstream LLM API applies record padding or token batching, does that protection cover the user-facing connection after an application frontend? NetEcho demonstrates that a content-aware frontend that consumes decoded upstream tokens and re-serializes responses into new chunks (Trace A token counts per group; Trace B character lengths) restores topic and conversation distinguishability [7]. SplitDecoy establishes the same boundary on a backward return path: even when forward split-model activations are obfuscated with decoys, the returned backward gradient exposes nonzero row support on every tested frame across nine seeds [8, Sections 5.1–5.3].

Remark 1 (target-oblivious postprocessing). Let S be the hidden target, Z the complete protected upstream representation and Y the downstream observation. If $S \to Z \to Y$ forms a Markov chain (any downstream framing or randomization is conditionally independent of S given Z), then by the data processing inequality [53, Theorem 2.8.1],

$$I(S;Y) \le I(S;Z).$$

A content-aware frontend that decrypts, strips padding and re-chunks plaintext ($S \to (Z, \text{plaintext}) \to Y$), or a backward pass that computes gradients from unprotected loss masks, need not satisfy $I(S;Y \mid Z) = 0$. The data processing inequality then no longer certifies the bound, and leakage must be checked separately. Section 8 reproduces the re-emission mechanism on TLS sockets and tests a streaming emitter.

5.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. A comparison that groups WRA's decoy requests [6] and AHB's humanized actions [9] together as "cover traffic" can create an apparent paradox: WRA reports a clean privacy improvement (one concurrent decoy reduces functional-intent reconstruction E_{func} from 0.770 to 0.610), whereas AHB reports severe task collapse (Trip Planning success falls from 0.75 Raw to 0.15 under combined `Online+Fake` humanization).

Binding their result records shows that WRA and AHB sit at opposite poles along *two structural axes*:

1) **Threat polarity inversion (Type I vs. Type III):** WRA is a Type I privacy defense protecting a user's research intent from a network/domain observer. AHB is a Type III bot-evasion mechanism designed to make an automated mobile GUI agent mimic human touch patterns so it bypasses platform anti-automation controls.
2) **Action-interface risk magnitude (discarded read-only queries vs. live GUI state mutations):** WRA's decoys are *read-only search and retrieval prompts* whose returned pages are discarded locally before synthesis; they never invoke state-mutating tools on the user's account. By contrast, AHB's `Fake` operator injects *unfiltered live GUI touch events* directly onto stateful mobile application screens, triggering unintended navigation and form submissions. The non-Fake `History+Long` configuration changes timing/trajectory history. These source-reported condition rates are not treated as a matched effect: condition-specific sample denominators could not be recovered from the source.

Dispatch-time authorization versus terminal-state scoring. Separating cover execution routes also exposes a

fundamental *analytical design argument* regarding agent benchmark evaluators: terminal state-snapshot scorers (such as τ-bench's native reward, which hashes the final database state against the reference state) are **path-oblivious**. If an unconfirmed live cover action happens to perform the exact state transition that the user's task later intended (or preempts a real action so the real action fails with a status error while leaving the terminal state identical to baseline), a terminal-snapshot scorer awards a perfect score of 1.0 although the endpoint does not establish authorization of the earlier call. Systems admitting cover to live dispatch must therefore enforce authorization and provenance checks at dispatch time and inspect tool state-machine transitions (such as one-way status flags or append-only audit ledgers, see Section 8).

5.3. Case 3: bind adaptive retraining (D_2) and cost objects

The $D_0 \to D_2$ rebound and component-cost boundaries. Selecting a defense operating point requires pairing an adaptive attacker evaluation (comparing a clean-trained classifier D_0 against a defended-trained classifier D_2 on the *exact same defended test set*) with the cost of the actual deployed mechanism. In UI Traces [5], retraining on timestamp-delayed archive traces (D_2) recovers substantial model-attribution accuracy (confirmed by our supplementary archive reanalysis on 1,135 test records: $+0.1503$ nats log-loss reduction over D_0, 98.333% Bonferroni-adjusted CI (three contrasts) $[0.1205, 0.1815]$; Supplement S4). Section 8 gives matched adaptation examples in which D_0 reads 50% while D_2 reads 100%, and shows why count rules and profiling decoders must accompany a learned observer.

On the cost side, record binding prevents cross-mechanism price substitution: Early Bird's 6ms/4% GPTCache semantic de-identification overhead does not price K-prefix KV-cache isolation [3]; KVGov's $1.3\,\mu$s HMAC key-derivation microcost omits workload-dependent lost cache reuse [15]; and SpliceDefense's $<$ 5ms/$<$ 1% QCP padding overhead prices chunk padding rather than combined CTBF+QCP deployment [30]. Conversely, CachePrune jointly binds marked-token attack suppression, QA/summarization utility and serving throughput on selective KV sharing [16], and OTRO bounds conditional access-count leakage ($3.94 \to 0.04$ bits given byte length) alongside separately scoped unpadded TTFT (4.59%) and epoch-padding (2.8ms) costs [17].

6. Comparability Audit Across All 58 Core Families

6.1. Unified 58-core disposition and 39-record ledger

Across all 58 core families in our unified collection (43 from the October 8 snapshot plus 15 from the October 21 expansion), every family has a verified intervention disposition (Supplement S5 (Table S11)): **20 families evaluate mitigation components** (16 snapshot + 4 additions: `wu2026cacheprune`, `lee2026otro`, `chen2026sparleak`, `hadad2026shape`), two use fixed guardrails, five evaluate interventions for other objectives or split-decoy boundaries (4 snapshot + `politis2026splitdecoy`), 22 report no mitigation evaluation within the inspected scope (13 snapshot + 9 additions), five are Type IV authorized diagnostic components (4 snapshot + `revankar2026mirage`), and four contain unbound quantitative defense assertions. The 20 evaluated-mitigation families and contextual comparators contribute **39 source-bound result records** (33 in the baseline snapshot plus 6 from the four October 21 mitigation additions) and **78 utility/cost field positions**. Table 8 organizes the eight intervention families by threat polarity (Types I–III), receiving view and deployment boundary.

6.2. Within-family and within-cohort record comparisons

While the machine-readable audit retains all $\binom{33}{2} = 528$ baseline pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by construction. The 13 within-family pairs are included in the 102 within-cohort pairs: $528 = 426 + 102$, with no additional subtraction of 13. Focusing on the **13 within-family pairs** (multiple records inside the same paper) and **102 within-cohort related pairs** (66 traffic-envelope, 28 cache-service, 3 behavioral-humanization, 3 generator-fingerprint, 1 local-memory, 1 fixed-instruction; Supplement S8) reveals why paper-level comparisons fail:

- Even within the **13 same-family pairs** (plus 2 same-family pairs in the additions: `CACHEPRUNE-selective` vs. `CACHEPRUNE-context` and `OTRO-unpadded` vs. `OTRO-epochpad`), nearly half (6/13 baseline; 7/15 full) measure different primitives, targets or cost objects (e.g., Early Bird's K-prefix oracle simulation vs. its GPTCache semantic prototype; SpliceDefense's CTBF signal suppression vs. QCP-only prefill cost; OTRO's unpadded TTFT vs. derived epoch-padding cost).
- Among the seven aligned-outcome same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture feature ablation at 0.17 accuracy vs. live loopback OHTTP at 0.30 accuracy) and three lack a verified matched-test binding, so none licenses cross-record subtraction as a single paired deployment effect.
- Across the **102 related-cohort pairs**, only 16 (15.7%) support a source-qualified descriptive comparison, whereas 80 (78.4%) exhibit positively established receiver, target or cost-object mismatches and 6 (5.9%) remain unresolved (the supplied 50-pair

TABLE 7. SOURCE-LOCAL FINDINGS, CROSS-STUDY SYNTHESIS AND THE RESULTING MEASUREMENT OBLIGATION. THESE ARE REUSE JUDGMENTS, NOT ACCUSATIONS THAT THE SOURCE AUTHORS CLAIMED END-TO-END PROTECTION.

Check	Source-local evidence and locator	Synthesis added by record separation	Changed judgment / controlled evidence
1. Receiver	NetEcho IV-C–E/Fig.5: frontend re-emission; SplitDecoy 5.1–5.3: backward gradient support.	Protected upstream and forward views do not cover newly exposed downstream or backward inputs.	Bind the final observer or prove the postprocessing condition. A buffered cap (2.18×) does not price a token stream: per-event padding leaves verbosity at 100% (E5).
2. Execution	WRA 8.1/Table8: read-only query decoys; AHB 5.2.3/Table1/E.3: live GUI humanization.	Objectives, live action semantics and protected state scopes differ, so their utility outcomes are not contradictory.	Evaluate task objects, full account and shared database separately. Same-item writes change status; randomized and decoy cover stay at 77–100% BA; dispatch cover leaks on the tool link (E6, E7).
3. Adaptation and cost	UI Traces 6.1: delayed-trace refit; Dangol 9.7: capture versus live protocol; Early Bird's KV-prefix versus semantic-cache mechanisms.	A paired attack contrast requires the same defended tests; costs from another mode or mechanism cannot price the claimed protection.	Keep source-local comparisons and mark missing bridges unresolved. D_0 reads 50% where D_2 reads 100% under batching and probabilistic cover (E5, E6).

subsample, not all 102 pairs, has 94.0% agreement across its 150 judgments, $\kappa = 0.916$).

6.3. Nine-item split reporting audit across 20 mitigation families

Table 9 reports nine audit items R1–R9, separate from deployment Checks 1–3. They are requirements for the particular reuse claims considered here, not universal obligations that every component paper must satisfy. R4 records protected-condition fitting; R5 records a matched clean-versus-adapted contrast on defended tests. R7–R9 record inference, utility and cost. A family-level Bound entry means at least one scoped result meets the item, not that every result in that family does.

For Dangol, R5 is Bound for in-fold stored-capture experiments; the live-refit paired comparison in Table 5 remains NR-V. These are different records, so we retain the family count of two and explicitly restrict its meaning.

SoK Takeaway 2 (Joint evaluation gap). Across 20 mitigation families, 17 bind inference outcomes, 11 bind cost, 3 bind utility and 2 bind a matched adaptation contrast. Excluding structurally inapplicable entries, the latter three rates are 11/19, 3/17 and 2/11. NR-V means unverified binding, not an absent experiment. CachePrune binds inference, utility and cost at one operating point; this is not a claim that it also binds adaptive evaluation.

7. Match Transfer Claims to the Generator

7.1. Different tests rule out different explanations

A fingerprint is generated by a model interacting with tasks, prompts, tools, browser state and serving infrastructure. A holdout establishes portability along the axis it changes: new tasks, browser configurations, network topologies and hardware are distinct deployment tests. Supplement S2 maps each dependency to its supporting holdout.

UI Traces fixes a browser/action harness while testing tasks and environments. FP-Agent combines behavior with static browser attributes; shared visitor proxies are not authenticated independent people or hosts [5], [18]. Task separation can remove memorization of those task instances while retaining configuration identifiers. This is not proof of an invalid original split; it bounds portability. The traffic benchmark separately tests generating topology, so its shift is a different test rather than a stronger score on the same estimand [42].

LeakyLMs illustrates the boundary for system-property rather than agent identity. Its deployment probe observes streaming events through an ordinary client API. Its architecture branch additionally assumes known GPU type, same-class calibration and single-GPU execution [56]. Remote architecture inference therefore does not establish hardware-agnostic inference. The joint search in its remote example ranks the true configuration 28/2068, distinct from a conditional parameter search or a near match; those outcomes must not be collapsed into universal exact recovery. Source-reported local architecture generalization and remote deployment signatures remain positive evidence under their own assumptions.

7.2. Representation increment is not portability

Within a fixed observer and test population, coarse counts versus sequence features test whether the latter improve a specified predictor. WRA already uses ordered domains before adding timing; Kang changes class sets across some conditions [6], [40]. These test different contrasts from an order-only effect. The supplementary archive example fixes labels and units and finds a 0.3447-nat sequence increment under archive-receiver ordering. Its small common-task control leaves task-distribution explanations unresolved (Supplement S4).

Adjacent coding-agent PR attribution depends on repositories, submission pathways and artifact generation. Its mixed-feature result does not isolate an auxiliary-only C3 view; inspected code computes weighted F1 and reduces features before cross-validation, but the exact paper-run

TABLE 8. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [7], [27].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains NR-R.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [25].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are NR-R.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [9].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [26], [15], [16].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median $1.3\,\mu$s prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [30].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and < 5ms/$< 1\%$ prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [54], [55], [17].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

TABLE 9. NINE-ITEM SPLIT REPORTING AUDIT OVER THE 16 ORIGINAL EVALUATED-MITIGATION FAMILIES (AND 20 TOTAL INCLUDING THE FOUR OCTOBER 21 MITIGATION ADDITIONS IN PARENTHESES). R1–R9 DESCRIBE EVIDENCE FOR SCOPED REUSE CLAIMS; THEY ARE SEPARATE FROM DEPLOYMENT CHECKS 1–3.

Audit item	Bound	Partial	NR-R	NR-V	NA
R1. Target & protection polarity	16 (20)	0 (0)	0 (0)	0 (0)	0 (0)
R2. Receiver fields & linkage	13 (17)	3 (3)	0 (0)	0 (0)	0 (0)
R3. Placement & action dispatch	13 (16)	2 (3)	0 (0)	1 (1)	0 (0)
R4. Protected-condition fit	4 (4)	1 (1)	0 (0)	4 (6)	7 (9)
R5. Matched adapted test	2 (2)	2 (2)	0 (0)	5 (7)	7 (9)
R6. Workloads & generator shifts	14 (18)	2 (2)	0 (0)	0 (0)	0 (0)
R7. Defended inference effect	14 (17)	0 (1)	2 (2)	0 (0)	0 (0)
R8. Task or component utility	2 (3)	2 (3)	10 (11)	0 (0)	2 (3)
R9. Mechanism or protocol cost	10 (11)	1 (3)	4 (5)	0 (0)	1 (1)

Bound: explicitly established for the scoped claim. Partial: a supported element coexists with a proxy or unlinked operating point (e.g., AHB refit code without verified Table 1 run binding; WRA Table 11 visibility utility separate from Tables 8–9 decoys). Only one family across the expanded 20-family mitigation set (`wu2026cacheprune`) jointly binds R7, R8 and R9 on the same operating point. Supplied two-column agreement over the 144 baseline cells is 93.1% ($\kappa = 0.876$, 95% CI $[0.798, 0.943]$).

binding remains unverified [10]. We neither rewrite its headline score nor quantify optimism. WildAuth is adjacent under the same rule; input/partition ambiguities do not supply a model-independent response-only privacy claim [11].

7.3. Resulting claim hierarchy

Task-specific identification, task-disjoint identification in a fixed harness, configuration/topology transfer and hardware-calibrated architecture inference require different evidence. Move between them by changing the relevant generator while preserving target labels and the receiver contract. For a representation increment, compare representations on identical units and disclose predictor/selection restrictions. This claim-to-holdout rule lets a reader reuse each author test for the shift it measures.

8. Controlled Transfer Tests

The experiments operationalize the audit's three bindings: observation, execution and matched evaluation. E5–E7 isolate mechanisms using saved outputs and reference programs; E9 evaluates real LLM-generated trajectories with training-only calibration. Seven author-written judgments were recorded before E5–E7; their reassessment is retained in Supplement S12. These cases test mechanisms, not the audit's external decision accuracy.

For E6/E7/E9, a cell meets the *evaluated-suite criterion* when maximum BA has an upper 95% bound below 0.60 and maximum AUROC is below 0.65. A lower BA bound above 0.60 indicates above-threshold leakage; other cells are inconclusive. We recompute the suite maximum within each of 2,000 common bootstrap draws: tasks for E6/E7 and customers for E9, retaining their tasks and repeats together. Intervals condition on fitted predictions and do not cover unknown attackers.

A byte-transparent relay captures post-handshake TLS application records on loopback connections. TLS 1.3 session tickets are disabled. E5–E8 use saved text or stand-in planners; E9 invokes a real model and executes its structured actions in the local public retail simulator. No production account or tool is modified.

8.1. Transport: buffered and streaming emitters

Buffered emitter. The corpus has one prompted brief and one prompted expanded `gpt-6-luna` response for each of 24 topics (12 training and 12 test topics); these are designed conditions, not natural length samples. In a two-link TLS 1.2 replay, a frontend that decodes and re-chunks the upstream text leaves the terminal link 100% separable when only the upstream link is padded; padding both links reduces the evaluated classifier to 50%. This reproduces NetEcho's mechanism [7]. With workload-matched denominators, 512-byte buckets cost 2.30× and 1.43× for two sentence-prefix workloads (T2, T3); a cap fitted on training lengths (1,280 bytes) yields 50% at 2.18× for verbosity (T1). Under offsets, 1024-byte buckets and random tails, refitting (D_2) gains 13.5–50.0 points over D_0; constant caps give no gain. Supplement S11 retains all cells.

E5: the same texts as a token stream. These figures describe an emitter that frames a response only after generation is complete. LLM APIs usually stream, and the streaming attacks in our corpus observe per-token records [1], [27]. E5 emits the same 48 texts as server-sent chunk events, one TLS record per token, over TLS 1.2 (AES-128-GCM) and TLS 1.3 (AES-256-GCM). Token boundaries use o200k_base as a proxy; the generating model's tokenizer and timing were not saved, so latency is reported in token intervals. The framings are per-event random padding of 1–64 bytes (an obfuscation field), a constant 144-byte event, 8-token batches padded to a constant size, and constant events plus dummy events up to the largest training token count. Besides D_0 and D_2 threshold and sequence classifiers, a closed-world attacker profiles all 48 candidate responses in an independent run and matches each observation to the nearest record-length sequence.

Table 10 resolves P1–P3. Per-event padding hides token lengths but not their number. Verbosity remains 100% separable for both attackers, and identification falls only from 100% to 58–62% (chance 2.1%). Batching sends fewer envelopes (0.21× bytes) and drives D_0 to 50%, while D_2 still scores 100%. A constant-shape stream reaches 50% at 2.42× stream bytes and doubles completion time (208 versus 105 token intervals); two test outputs overflow the fitted count and remain identifiable. The buffered cap's 2.19× is a different cost object: its first byte arrives only after the full response. The 2.18× result therefore does not price a streaming deployment; the result illustrates the emitter and cost bindings in Check 1 and R9.

TABLE 10. E5: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				T1 acc.		T3	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Tokens: o200k_base proxy. Count cap: 208 (T1) and 88 (T3) events fitted on training outputs; 2/24 and 5/24 test outputs overflow it.

8.2. Execution: state scope and randomized cover

State scope. We replay the 115 public retail reference programs. Same-item writes change workflow state: the item-modification tool sets the status to `pending (item modified)` and appends a refund entry when item and price are unchanged (173/291 pre-insertion units). A no-change requirement $f(s, a) = s$ thus differs from repeated-call idempotence $f(f(s, a), a) = f(s, a)$. Cancelling another simulated user's order preserves the active account in every unit while changing the shared database in every unit. One cover call at a known first or last position is removed by a position or count rule with 100% accuracy. These are properties of one simulator and of a deliberately weak cover design (Supplement S11).

E6: randomized cover against policy-aware attackers. E6 asks what a properly randomized cover achieves in the same tool-dispatch view, which shows ordered tool names and response lengths, including failures. Cover calls are drawn from other users' reference programs, inserted at uniformly random positions, and executed in a per-session sandbox (shadow) or, as a control, live. The policies add one cancellation with probability 0.5 or 1, pad the cancellation count to two, interleave one or two complete decoy programs, or equalize the program *shape*. The last pads every tool's count to its maximum over the public program population, a label-free but full-population template. Decoy pools and framing caps also use all 115 tasks. This is known-population defense calibration; attacker holdout does not make it unseen-task defense evaluation. Attackers are

gradient boosting and logistic regression over tool counts, bigrams and response lengths, fitted on clean (D_0) or defended (D_2) sessions, plus a count rule. We use task-grouped 5-fold cross-validation over all 115 tasks and three seeds, with two targets: cancellation (18 tasks) and return (32 tasks).

The dispatch columns of Table 11 show that adding one cancellation always takes D_0 to 50% BA, but D_2 recovers 100% (P2). Equalizing the cancellation count defeats the count rule (50%), whereas a learned model using correlated tools reaches 76.9% (P4). Decoy programs, the read-only defense pattern of WRA [6], leave 83.5–91.9% because most decoy programs contain no cancellation. Shape equalization gives 50.0% and 54.5% point estimates, at 6.7 cover calls per real call. Sandboxed cover keeps database equality at 100%; the same cancellation executed live yields 0%. Equality verifies state preservation under the isolation design, not task quality, authorization or deployment utility.

8.3. E7: composing execution and transport on the agent–tool link

Figure 1 draws observation and execution as separate paths; a remote tool server joins them on one encrypted connection. E7 replays four cover policies over a TLS 1.2 agent–tool link with one length-prefixed request and response per call (4,140 sessions). Framing is raw, 1024-byte buckets or constant (512-byte requests; 3,072-byte responses, the 95th percentile, with larger responses padded to 1024-byte multiples). The network observer sees direction-tagged record lengths and no tool names. In addition to the learned models, a profiling attacker learns a size-to-tool table by calling the public tool API on training-fold calls, decodes test traces and applies the dispatch attacks.

Under raw framing, request-only identification is 89.9% with cancellation-stratified folds and 91.4% with return-stratified folds (582 calls; training-majority baseline 29.4%). The downstream decoder additionally uses response sizes and is fitted on training-fold calls. Count and decoy cover leave 72.7–95.7% BA on the network link (P5). They already fail in the dispatch view, so this is persistence of leakage, not a protected-source-to-unprotected-target transition.

Framing alone leaves bucketed return inference at 73.9%; other framing-only cells are inconclusive. Shape cover with raw framing has cancellation BA 51.2% but return BA 60.5%; with constant framing the values are 54.4% and 50.0%. Thus several point estimates are near chance. No network configuration meets the evaluated-suite criterion for *both* targets. The joint constant-framing configuration costs 26.3× bytes and 7.7× calls, with database equality 100%; this is one measured operating point, not a minimum necessary cost (P7). Supplement S12 gives all suite-maximum intervals.

E6/E7 calibrate on the known program population; E8's stand-in planners (Supplement S13) motivate the online, training-only dispatcher that E9 evaluates with real model decisions.

8.4. E9: real agents and training-only defense calibration

E9 tests defense reuse on autonomous model-generated trajectories under a strict calibration boundary. GPT-6.1 Sol (low reasoning effort) plans and responds from the retail policy, public tool schemas and actual sandbox observations. A separate context of the same model simulates the user. The rules arm adds one customer-details read before order access and a read before each order write. All 115 tasks have two prompt arms and three repeats, completing the frozen 690-run protocol. All 414 training runs on 69 tasks precede freezing of a shared count template, donor pools and framing caps; the 276 test runs cover 46 tasks from 24 disjoint customers.

The online dispatcher inserts training-derived cover on shadow state using only past calls and flushes unmet counts at episode end. Real calls pass through unchanged. Four program conditions per prompt isolate raw transport, padding, shape cover and their combination. Training-fitted attackers evaluate dispatch and network observations across 2,760 TLS replay sessions. Every test task, failure and handoff remains in the analysis. Supplement S14 supplies the full ablations and customer-cluster intervals.

Compliant workflows still reveal intent. Table 12 shows baseline network BA of 97.2% for cancellation and 94.0% for return. Rules raise measured workflow compliance from 86.2% to 97.1%, yet network BA remains 97.2%/93.5%. Task reward is 110/138 (79.7%) for baseline and 105/138 (76.1%) for rules; the paired difference is -3.6 points (95% interval $[-9.5, 1.4]$). Thus workflow rules successfully alter measured compliance while leaving a strong intent signal in the encrypted trace.

Joint shaping removes most measured inference advantage. On baseline trajectories, online shadow cover with constant framing reduces network BA by 47.2 points for cancellation and 44.0 points for return, reaching 50.0% for both at 15.4× bytes and 3.8× calls. The ablations identify the value of combining the mechanisms: padding alone leaves cancellation/return BA at 66.1%/55.5%, and shape cover alone at 60.8%/51.9%.

Its 95% upper bounds are 60.6%/65.5%, and rules plus shaping yields 50.0%/60.6% (difference interval spanning zero, S14). Shadow calls never enter the planner's context, so the defense is planner-transparent: each trajectory and its reward are preserved by design, and its online cost appears as extra calls.

8.5. E10–E12: power, operating point and mechanism

E10: cross-fitting. The E9 test split has only five to seven positive customers per target. E10 therefore reruns the frozen pipeline, without new model calls, under customer-grouped stratified 5-fold cross-fitting over all 690 runs (115 tasks, 53 customers). In every fold, the template, donor pools, caps and all attackers are fit on the other folds'

TABLE 11. E6/E7: RANDOMIZED COVER ON 115 RETAIL REFERENCE PROGRAMS × 3 SEEDS. ENTRIES ARE THE BEST HELD-OUT BALANCED ACCURACY (%) OVER ALL ATTACKERS FOR THE CANCELLATION (C, 18 POSITIVE TASKS) AND RETURN (R, 32) TARGETS. DISPATCH VIEW: TOOL NAMES AND RESPONSE LENGTHS. NETWORK VIEW: TLS 1.2 RECORD LENGTHS ON THE AGENT–TOOL LINK UNDER RAW, 1024-BYTE BUCKET OR CONSTANT FRAMING.

	Dispatch view (E6)				Network view (E7)						
					Raw		Bucket		Constant		
Cover policy	C	R	Calls+	DB eq.	C	R	C	R	C	R	Bytes×
No cover	100.0	100.0	0.00	100.0	100.0	95.7	60.1	73.9	59.7	53.6	3.4
Cancel, p=0.5	86.1	100.0	0.10	100.0	—	—	—	—	—	—	—
Cancel, p=1	100.0	100.0	0.20	100.0	—	—	—	—	—	—	—
Cancel, p=1, live execution	100.0	100.0	0.20	0.0	—	—	—	—	—	—	—
Cancel count → 2	76.9	100.0	0.35	100.0	72.7	95.7	67.5	66.0	64.9	59.1	4.6
One decoy program	91.9	85.9	1.01	100.0	91.9	85.9	60.0	59.1	59.7	56.8	6.9
Two decoy programs	83.5	78.1	1.98	100.0	—	—	—	—	—	—	—
Shape equalization	50.0	54.5	6.71	100.0	51.2	60.5	55.3	50.6	54.4	50.0	26.3

Calls+: cover calls per real call. DB eq.: orders/users equal to reference. Cover is sampled from other users' reference programs and placed at random positions; all policies except "live" run cover in a per-session sandbox. Attackers: D_0/D_2 gradient boosting and logistic regression, a count rule and, on the network view, a size-profiling decoder; task-grouped 5-fold cross-validation. Bytes×: constant framing relative to raw framing without cover. Suite-maximum intervals: Supplement S12. Defense templates/pools/caps use all 115 tasks; attacker fitting uses training folds. No network configuration meets the criterion for both targets.

TABLE 12. E9: COMPLETE REAL-AGENT EVALUATION WITH TRAINING-ONLY DEFENSE CALIBRATION. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FIXED ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Reward (%)	Bytes×	Calls×
Baseline	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	79.7	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	79.7	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	79.7	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	76.1	15.72	3.88

Joint = online shadow cover plus constant framing. Padding and shape ablations use baseline trajectories. Ratios use raw transport within the same prompt. Program conditions reuse each trajectory, preserving reward by construction. Network measurements are TLS 1.2 loopback replay of actual model-generated calls; 414 training runs preceded defense freezing and test collection. Full methods and ablations: Supplement S14.

customers only; out-of-fold predictions are pooled. Positive customers rise to 14 (cancellation) and 18 (return). An analytic TLS 1.2 record model reproduces all 2,760 socket captures and 120 fresh ones exactly, and the pipeline reproduces every frozen-split cell. Table 13 shows undefended leakage of 91.7%/85.5%. The E9 operating point lowers it by 31.2 and 33.4 points (paired intervals $[-41.6, -21.5]$ and $[-41.0, -21.5]$). Rules do not interact with the defense: with bucketed shaping, the rules-minus-baseline difference is $+1.2$ and -2.0 points, and both intervals span zero. Adding random-forest, k-NN and forest size-decoding attackers changes no E9-point or bucket cell by more than 1.7 points (Supplement S15).

E11: operating points. Six configurations vary cover scope and framing. Replacing the 95th-percentile constant response cap by 1024-byte buckets lowers cost from 19.8× to 12.4× bytes and reduces leakage to 52.2%/52.7% (paired reductions 39.4 and 32.8 points, AUROC ≤ 0.53). Its upper bounds of 60.5%/60.2% sit at the 60% criterion. With the rules prompt, return inference meets the criterion (50.7%, upper bound 57.3%). A rule fixed in advance selects, on training folds alone, the cheapest configuration whose inner BAs are all at most 60%; it chooses the bucket point, which on the frozen 46-task test costs 9.8× rather than 15.4× bytes at 53.2%/58.4%.

TABLE 13. E10–E12: NETWORK-VIEW MAXIMUM BA (%) OVER THE FIXED SUITE WITH CUSTOMER-CLUSTER 95% INTERVALS. CROSS-FITTING RAISES POSITIVE CUSTOMERS TO 14 (CANCEL) AND 18 (RETURN). BASELINE PROMPT UNLESS STATED; BYTES/CALLS RELATIVE TO RAW TRANSPORT. WRITES: STATE-CHANGING CALLS.

Configuration	Cancel BA	Return BA	B×/C×
E10: cross-fitted, 690 episodes, 53 customers			
Undefended	91.7 [82.1, 100.0]	85.5 [76.6, 93.4]	1.0/1.0
E9 point (constant)	60.5 [51.5, 70.5]	52.1 [50.0, 60.3]	19.8/5.0
1024-B buckets	52.2 [50.0, 60.5]	52.7 [50.0, 60.2]	12.4/5.0
Writes-only cover	53.5 [50.0, 65.8]	56.1 [50.0, 67.4]	11.3/2.9
Rules + buckets	53.4 [50.0, 60.6]	50.7 [50.0, 57.3]	12.4/5.0
E11: chosen on training folds; frozen 46-task test			
E9 point	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	15.4/3.8
Selected: 1024-B buckets	53.2 [50.0, 69.5]	58.4 [52.9, 69.1]	9.8/3.8
E12: undefended observer ablation, cross-fitted			
Writes removed	55.2 [50.0, 68.7]	59.9 [50.5, 71.7]	–
Writes only	91.7 [82.1, 100.0]	84.9 [78.2, 93.7]	–

E12: where real-agent leakage lives. An oracle ablation of undefended traces explains the rules result. State-changing calls alone support 91.7%/84.9% network BA; removing them leaves 55.2%/59.9%. The rules prompt reduces distinct read-only call sequences from 74 to 62 but leaves state-changing sequences unchanged (37 versus 36, 4.08 versus 4.09 bits). Shaping only writes is nonetheless insufficient (53.5%/56.1%; no writes-only configuration passes inner-fold selection): effective shaping must cover the full call profile and its sizes.

8.6. What the cases establish

Each experiment instantiates one audit obligation and changes a reuse judgment. E5 shows that the emitter fixes both the leaking count and the cost object. E6/E7 show that refitted attackers and encrypted size decoding recover tool intent that clean-trained or dispatch-only evaluations miss. E9–E12 carry these bindings to real agents: leakage concentrates in state-changing calls, behavioral compliance does not reach it, and training-only program shaping removes most of it at a measured, tunable cost.

SoK Takeaway 3 (Bind the projection the emitter produces). Padding applies to the record sequence it equalizes. Buffered caps, per-event padding and batching have distinct leakage and cost objects. Evaluate the attacker on defended data and price the same emitter that supplies the protection claim.

SoK Takeaway 4 (Shape what the agent changes, and price it). Across 690 real-agent runs, rules yield 97.1% measured compliance while intent inference remains above 93%, because intent leaks through state-changing calls that the rules leave untouched. Training-calibrated joint shaping lowers inference by 31–47 points; bucketed framing reaches 52–53% at 12.4× bytes, 37% below constant framing. Evaluate behavioral compliance, state preservation, metadata protection and cost as distinct outcomes.

9. Implications, Ethics and Limits

9.1. Reporting obligations and open problems

The nine audit items specify what makes defense results comparable: R1 target and protected interest; R2 receiver fields and linkage; R3 intervention placement and execution; R4 defended-condition fitting; R5 matched adaptation; R6 workload and generator shifts; and R7–R9 inference, utility and cost. A component study can satisfy its own scope without implementing a complete autonomous agent; reuse across components requires the missing bindings.

Our measurements motivate three priorities. Streaming defenses should report the record sequence they equalize together with byte, completion-time and first-byte costs. Tool contracts should specify protected state and the effects of successful and rejected calls. Agent defenses should shape the full call profile, starting from state-changing calls, fit parameters on training workloads, test adaptive attackers on every exposed link, and report the leakage–cost frontier rather than one operating point. Workload-level cache-reuse costs and cross-generator fingerprint portability remain separate problems [15], [16].

9.2. Ethics and artifact scope

Tool executions and traffic captures use cloned public benchmark databases and local loopback sockets. E9 sends public benchmark instructions and simulated interactions to the model service; it modifies no production tool or real account. Shadow cover assumes a cooperating provider that does not commit its effects. Other-account cancellation is a simulator control, not a deployable decoy without controlled resources and an isolation argument. The corpus also covers bot-detection evasion; its authorization and objective differ from confidentiality protection [9], [6].

The artifact preserves sources, raw observations, structured model decisions, frozen calibration and executable checks. AI-assisted implementation and manuscript revision do not constitute independent validation. The paper distinguishes saved-output replays, real model-generated trajectories and local TLS measurement.

9.3. Threats to validity

The purposive corpus and query overlaps do not estimate field-wide recall. Supplied coding columns support agreement calculations, but coder independence and adjudication are not established from those columns. The 24 author-constructed scenarios illustrate audit rules; external decision accuracy, incremental value over a simple checklist, and independent screening of the 617-record second search remain open.

E5 uses 48 saved outputs and a proxy tokenizer; E6/E7 use reference programs with known-population calibration. E9–E12 cover one public retail environment and one mutable model alias, also used for user simulation; earlier development had access to all 115 tasks. Cross-fitting adds power, not a new population, and intervals condition on fitted attackers. Program conditions reuse trajectories, so equal reward is structural. External replication, broader model/environment transfer, provider-traffic measurement and interactive latency/feedback effects are the next evaluation targets.

10. Conclusion

This SoK connects 82 study families and 39 scoped result records to an executable evaluation of LLM and agent side-channel defenses. The audit identifies which observation, execution and evaluation bindings govern a result's reuse; controlled experiments demonstrate the leakage and cost consequences of changing those bindings.

The complete 690-run real-agent study provides a concrete deployment lesson. Workflow rules achieve 97.1% measured compliance while intent inference exceeds 93%. Training-calibrated online shaping reduces baseline network BA by 47.2/44.0 points to 50% at 15.4× bytes and 3.8× calls. Intent lives in state-changing calls that rules leave untouched; bucketed framing cuts the byte cost by 37% at matching or lower leakage.

Appendix A. Reproducibility and Notation

The accompanying *Supplementary Evidence* and package directory provide the following modules: S1 historical search records; S2 corpus/prior-work maps; S3 retained pilots; S4 public-trace reanalysis; S5 intervention inventory; S6 original conclusion sensitivity; S7 workflow comparison diagnostics and the 24-case retrospective validity benchmark; S8 the pairwise and nine-item split audit (with supplied coding records); S9 controlled mechanism demonstrations and extensions (E1 length-sensitivity gradient, E2 synthetic plans, E3 D_0-vs-D_2 adaptive attacker, and T2c 115-task / 97-eligible-task retail baseline and realistic cover policies); S10 the completed 565-record search update and exact $60 \rightarrow 82$ reconciliation; the current replay/analysis artifacts in `experiments/reviewer-audit/`; and the second-round transfer tests, predictions and raw captures in `experiments/sp2027-r2/` (S11 retains the first-round controlled tables; S12 reports E5–E7 in full; S13 documents the agent pipeline). Earlier summaries in `experiments/revision-m1-m8/` are kept for provenance; current E6/E7 intervals, held-out identification and judgment reassessment use the post-review `experiments/sp2027-r3/` outputs. S12 reports the corrected E6/E7 summaries; R2 registration and historical outputs remain unchanged. S14 reports the complete 690-run E9 evaluation: 414 training and 276 test runs, training-only calibration, TLS captures and executable validation. `experiments/sp2027-r4-full-20261023/` reconstructs the full cohort from the immutable 663-run snapshot in `experiments/sp2027-r4-snapshot-20261023/` plus 27 additional runs. The earlier manuscript and snapshot remain in the archived release; the current paper uses the completed cohort throughout. S15 adds E10–E12 (cross-fitting, operating-point selection and leakage localization) in `experiments/sp2027-r5-crossfit/`, which verifies exact reproduction of every frozen-split cell.

Executable acquisition record. Table 14 records the actual historical IEEE/ACM native queries (October 8–9, 2026). S10 prints the four Semantic Scholar bulk queries (`S1R`, `S2`, `S3`, `S4`) and two native arXiv queries (`ARXIV-A1`, `ARXIV-A2`) completed on October 21, 2026 (615 occurrences, 565 unique records), together with the pre-audit and post-audit reconciliation table (Table 2).

Complete indexed-family citation inventory. The following citation groups enumerate all 82 indexed families. Categories are core eligibility and origin, not mutually exclusive threat objectives. The machine-readable index binds each key to its source version and locators. Core (58 families): [4], [5], [6], [18], [1], [2], [7], [3], [29], [27], [31], [30], [28], [57], [55], [58], [32], [59], [40], [25], [60], [38], [42], [61], [62], [54], [63], [20], [64], [65], [66], [67], [68], [69], [70], [26], [71], [9], [21], [22], [56], [15], [23], [45], [44], [43], [19], [41], [24], [39], [37], [8], [35], [36], [33], [34], [17], [16].

Adjacent (24 families): [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [10], [11], [82], [83], [12], [13], [14], [48], [84], [85], [86], [87], [88], [89].

TABLE 14. EXECUTED NATIVE QUERIES. S1 RETAINS THE FULL EXPORT ACCOUNTING; Q4 WAS ISSUED ONLY ON IEEE AND NOT EXPORTED IN THE ORIGINAL ACQUISITION.

ID	Exact query
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`

TABLE 15. READING KEY FOR THE SUPPLIED RECORDS. PROSE USES FULL MEANINGS; CODES SUPPORT JOINS.

Notation	Meaning
SCOPE	Secret/target, Channel, Observer, Pipeline, Exploitation goal.
Types I–IV	Confidentiality, Provenance attribution, Bot-detection evasion, Authorized diagnostics.
Origin A/B	Workflow-origin (A) / runtime-and-application-inherited (B) signal origins.
C1–C5	Five core eligibility conditions in Section 2.
P1–P8	Hardware, serving, transport, application, orchestration, tools, environment, multi-agent communication.
D0 / D2	Attacker fitted on clean / defended training data; compared on the same defended test set.
NR-R / NR-A	Unreported in inspected scope / unavailable material.
NR-V / NA	Unverified binding / structurally inapplicable.

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